Chapter 6 · Continuous distributions

6.0 Diagnostic

Unlabelled recognition across continuous models.

Identify the random object, distribution and region before calculating.

6.1 Continuous Probability Foundations

Probability is area; an exact point has probability zero.

For a continuous random variable, probability is area under its density. Therefore P(X=c)=0, and < versus ≤ does not change the probability.

6.2 Normal Distribution

Notation, parameters, standardisation and shape.

$X \sim N(\mu,\sigma^2)$. Changing μ shifts the curve; changing σ changes spread. Standardise with $Z=(X-\mu)/\sigma$.

6.3 Normal Regions and Forward Probability

Words → event → region → method → calculation.

Left: Φ(z). Right: 1−Φ(z). Between: Φ(zupper)−Φ(zlower). Draw or identify the region first.

6.4 Inverse Normal and Percentiles

Probability is known; the boundary is unknown.

Convert the requested region to cumulative-left probability, obtain z with NORM.S.INV, then $x=\mu+z\sigma$. NORM.INV can return x directly.

6.5 Evaluation of Normality

Integrate numerical and graphical evidence.

Compare mean/median/mode, empirical-rule proportions, skewness/kurtosis and suitable plots. No single check proves normality.

6.6 Excel Normality Laboratory

Use the official Steel data and supplied workbooks.

Progress VIEW → COMPLETE → RECONSTRUCT → EXECUTE FROM QUESTION. Sort X; create j; calculate $\alpha j=j/(n+1)$; compute Z with NORM.S.INV; scatterplot X on y against Z on x; integrate with histogram and descriptive evidence. Statistical reasoning and Excel execution remain separate evidence.

6.7 Uniform Distribution

Constant density between a and b; probability is rectangular area.

$f(x)=1/(b-a)$ on [a,b]. $P(c<X<d)=(d-c)/(b-a)$, $\mu=(a+b)/2$, $\sigma=\sqrt{[(b-a)^2/12]}$.

6.8 Exponential Distribution

Waiting time with rate and time units aligned.

$X \sim \exp(\lambda)$, $E(X)=1/\lambda$, $P(X\le x)=1-e^{(-\lambda x)}$, $P(X\ge x)=e^{(-\lambda x)}$. Poisson counts; exponential waits.

6.9 Inverse Exponential

Solve for a waiting-time boundary.

For cumulative probability α, $x_\alpha=-\ln(1-\alpha)/\lambda$. This lecturer-supported extension must not be confused with inverse normal.

6.10 Normal Approximation to Binomial

Check conditions, set centre/spread, then continuity-correct.

Use only when $n\pi>5$ and $n(1-\pi)>5$. Approximate centre nπ, SD $\sqrt{[n\pi(1-\pi)]}$. Exact x becomes $x-0.5<Xa<x+0.5$; ≤x becomes <x+0.5; ≥x becomes >x−0.5.

6.R Recognition & Selection Lab

Mixed stems without calculation labels.

Select object, model, formula family and region before calculation.

6.M Mixed Chapter Set

Independent interleaved practice.

No subsection cue; explain method choice.

6.B Chapter Final Boss

Altered, unlabelled transfer.

Completion alone does not establish mastery.

6.X Repair

Error-specific contrast and retest.

Immediate, delayed and transfer retests are scheduled from evidence.

Chapter 7 · Sampling distributions

> **SOURCE_READY · CURRENT CORE** — §2.6 verifies Student-t/T.DIST. **C7.52-C7.58 FPC remain PROVISIONAL / legacy-supported.**

7.0 Diagnostic

Object-first unlabelled diagnostic.

Ask what is random before selecting a distribution.

7.1 Population → Sample → Statistic → Sampling Distribution

A statistic varies across possible random samples.

A sampling distribution is the distribution of the calculated statistic over all possible samples of fixed size—not the distribution of raw observations in one sample.

7.2 Exact Sampling Distributions and Unbiasedness

Enumerate tiny populations to see long-run centre.

Each possible sample produces a statistic. The distribution of these statistics can show $E(\bar{X})=\mu$ even though one realised $\bar{X}$ need not equal μ.

7.3 Standard Error

The spread of a statistic across samples.

$\sigma_{\bar{X}}=\sigma/\sqrt{n}$. Increasing n narrows the sampling distribution without changing its centre.

7.4 X versus $\bar{X}$

One observation uses σ; an average uses $\sigma/\sqrt{n}$.

Underline the random object. X is one individual observation. $\bar{X}$ is an average of n observations and uses standard error.

7.5 $\bar{X}$ — σ known / Z

Normal sample-mean probabilities with known population σ.

$\bar{X} \sim N(\mu,\sigma^2/n)$, $Z=(\bar{X}-\mu)/(\sigma/\sqrt{n})$. Use cumulative-left area, complements and inverse functions as in Chapter 6.

7.6 Central Limit Theorem

The sampling distribution of $\bar{X}$ becomes approximately normal for large enough n.

The original population does not become normal. If the population is normal, $\bar{X}$ is normal for every valid n; otherwise use the source-aligned “large enough” principle.

7.7 $\bar{X}$ — σ unknown / Student t

Current 2026 source §2.6: sample-mean Student-t route when population sigma is unknown.

SDS Chapter 7 NOTES.pdf §2.6, p. 7: Xbar ~ t_(n-1); T=(Xbar-mu)/(s/sqrt(n)); df=n-1; Excel =T.DIST(tcalc,df,TRUE).

7.8 Sample Proportions

Sample proportion p: theoretical/estimated SE, forward probabilities, between limits and inverse central probability bounds.

C7.40 solves a p probability; C7.41 solves a between-limits p probability; C7.42 finds central probability bounds for p; C7.43 explains the effect of larger n.

7.9 Difference Between Two Independent Means

Difference between two independent sample means with canonical order, centre, combined-variance SE, event translation, calculation and n-effect.

C7.44 object; C7.45 order; C7.46 centre; C7.47 combined-variance SE; C7.48 >0 event; C7.49 >c event; C7.50 standardise/calculate; C7.51 effect of n.

7.10 Finite Population Correction

Scope-flagged bridge to Chapter 8.

PROVISIONAL: for sampling without replacement, inspect n/N; where source scope applies, multiply SE by $\sqrt{[(N-n)/(N-1)]}$.

7.11 Mixed Object & Method Selection

Select X, $\bar{X}$, p or $\bar{X}_1 - \bar{X}_2$ before method.

Then branch known σ → Z; unknown σ → t; proportion → conditions; two means → preserve order.

7.R Recognition & Selection Lab

Production before calculation.

State object, model, formula family and reason.

7.M Mixed Chapter Set

Independent familiar mixture.

Labels are removed.

7.B Chapter Final Boss

Altered and unlabelled transfer.

A pass adds evidence but is not sufficient by itself.

7.X Repair

Contrast the nearest competing routes.

Meaningful errors return for delayed and transfer retest.

Chapter 8 · Confidence interval estimation

> **PROVISIONAL — CURRENT 2026 SOURCE NOT YET ADDED**

8.0 Source Warning + Diagnostic

PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Diagnostic results guide provisional learning only and cannot satisfy final A1S2 readiness.

8.1 Point vs Interval Estimation

A point estimate is one statistic; an interval adds a quantified uncertainty range.

Match Xbar to mu and p to pi.

8.2 Estimator Characteristics

Unbiasedness, consistency and relative efficiency are different properties.

Unbiased: expected value equals parameter. Consistent: approaches the parameter as n grows. Relatively efficient: smaller variance among unbiased estimators.

8.3 Confidence, Alpha, Critical Value, Standard Error, Margin of Error

Estimate +/- critical value x standard error.

Confidence describes the long-run interval procedure. A fixed interval either contains the parameter or does not; it is not an interval for individual observations.

8.4 CI for Mu - Sigma Known

Known population sigma selects z.

OFF23 rule: if the population is not normal, use a large sample n > 30. Do not use the rhetorical p. 26 slide as a method-selection rule. Current 2026 Transcript 10 also supports the known-sigma → Z route.

8.5 CI for Mu - Sigma Unknown / Student t

Unknown sigma selects sample s, Student t and df=n-1.

Non-listed t-table row convention remains AWAIT_MEMO_OR_CURRENT_SOURCE. Exact independently calculated t values may check arithmetic but are not an official lecturer table convention. Current 2026 Transcript 10 also supports the unknown-sigma → Student-t route.

8.6 CI for Population Proportion

Use p +/- z*sqrt[p(1-p)/n].

Preserve strict source conditions np > 5 and n(1-p) > 5. Current 2026 Transcripts 10–11 support the sample-proportion centre, SE, approximation checks and confidence-interval structure.

8.7 Required Sample Size - Mean

Use n=z^2 sigma^2/e^2 and always round upward.

C8.42 is OFF23-CONFIRMED: use an expected conservative sigma or estimate sigma with pilot sample s when sigma is unavailable. Current 2026 Transcript 11 supports mean sample-size planning and the margin/confidence-width trade-off.

8.8 Required Sample Size - Proportion

Use prior/pilot p where available; otherwise use p=.5; always round upward.

C8.45 is OFF23-CONFIRMED, not 2026-confirmed. Current 2026 Transcript 11 supports proportion sample-size planning and conservative p=.5 when no estimate is supplied.

8.9 Finite Population Correction

For finite-population sampling without replacement, inspect n/N.

Use FPC=sqrt[(N-n)/(N-1)] when n/N > .05. Initial n0 adjusts to n=n0N/[n0+(N-1)].

8.10 Bootstrapping

Population -> original random sample -> bootstrap resample from observed sample -> with replacement -> same n -> statistic -> repeat -> order -> cut tails -> interval.

The 90%/100-resample worked example gives 36.6296 and 51.5926. All additional bootstrap practice is VERIFIED_ADAPTATION.

8.H Auditing / Totals - HOLD · HOLD

HOLD - A1 SCOPE UNCONFIRMED

Reference only. Excluded from DO THIS NOW, Learn routing, diagnostic, mastery, retrieval, repair, mixed sets, simulation, Final Boss and readiness.

8.R Repair

Error-specific provisional repair.

Repair tasks remain provisional evidence.

8.M Provisional Mixed Practice

Chapter 8-only and optional Ch6-8 provisional mixed discrimination.

coverageStatus=PROVISIONAL; cannot satisfy representative full-A1 readiness.

8.B Provisional Final Boss

Altered, unlabelled provisional transfer.

Cannot satisfy final 2026 source gate.

8.X Chapter 8 Source / QA

Source metadata, answer register and content QA.

Chapter status remains SOURCE_PARTIAL.

Chapter 9 · Fundamentals of Hypothesis Testing: One-Sample Tests

9.1 Hypothesis Foundations

Claims, H0/H1, the assumption under H0 and correct decision language.

A hypothesis is about a population parameter. H0 carries equality and is assumed true at the start; H1 is mutually exclusive. Conclude reject H0 or do not reject H0 — never accept H0.

9.2 Decision Risk: α, β and Power

Type I/II errors, significance, confidence coefficient and power.

α=P(Type I error); β=P(Type II error); confidence coefficient=1−α; power=1−β. For fixed n, lowering α increases β. Greater true-vs-null distance and larger n increase power; larger variance decreases it.

9.3 Tails, α and Rejection Regions

Let H1 determine the tail before calculating.

≠ gives two tails, < gives lower tail, > gives upper tail. A two-tail test splits α/2 into each tail; a one-tail test places all α in the directed tail.

9.4 Mean Test — σ Known: Z Critical-Value Method

Known population σ selects the Z route.

Use $Z=(\bar{X}-\mu_0)/(\sigma/\sqrt{n})$, determine critical value(s) from α and the tail, then compare Zcalc with the rejection region and conclude in context.

9.5 Z Test — p-Value + Excel

Interpret the p-value under H0, know its source synonyms, reconstruct the five-step p-value process, and compare p with α.

The p-value is the probability, assuming H0 is true, of a test statistic equal to or more extreme than observed. The deck also calls it the observed level of significance and the smallest α at which H0 could be rejected. Source workflow: H0/H1 → choose α and n → select statistic/distribution → collect data and compute statistic+p → decide and conclude in context.

9.6 Two-Tail Z and t Tests ↔ Confidence Intervals

Use the same-level CI as a decision cross-check for two-tail Z and Student-t tests.

At the same α/confidence pairing, a two-tail Z or t test rejects H0 when μ0 is outside the corresponding (1−α) confidence interval and does not reject when μ0 lies inside, provided the same model/conditions are used.

9.7 Mean Test — σ Unknown: Student t

Unknown σ selects Student t with df=n−1; normality evidence is checked explicitly.

Use $T=(\bar{X}-\mu_0)/(s/\sqrt{n})$, df=n−1. The deck notes that σ is rarely known in real business settings. For the t route, compare sample statistics with normal properties, inspect histogram/stem-and-leaf/boxplot symmetry, and inspect approximate linearity of the normal probability plot; then use T.INV/T.DIST/T.DIST.2T as appropriate.

9.8 One-Tail Mean Tests

Direction lives in H1; SDS188 uses equality in H0 for the one-tail teaching convention.

Lower: H0:μ=μ0, H1:μ<μ0. Upper: H0:μ=μ0, H1:μ>μ0. Put all α in the directed tail and use the matching Z/t probability.

9.9 One-Proportion Z Tests

Binary categorical data → p → conditions → Z → decision.

p=x/n. Check $n\pi_0\geq 5$ and $n(1-\pi_0)\geq 5$. Then use $Z=(p-\pi_0)/\sqrt{[\pi_0(1-\pi_0)/n]}$. Since x=np, the deck also gives the equivalent count form $Z=(x-n\pi_0)/\sqrt{[n\pi_0(1-\pi_0)]}$. Use the critical-value or p-value approach.

9.10 Power, Planning and Statistical Responsibility

Practical meaning, ethical planning, complete β/power computation and sensitivity to α/effect size.

Large n can make tiny effects statistically significant; small n can miss important effects. Plan the goal, hypotheses, tail, random sample, data type, α, sample size/power, procedure and contextual conclusion. Ethical issues named by the deck include human subjects, data collection, test choice, significance level, cleaning/discarding data and failure to report pertinent findings. Tutorial 9 Q21 is covered at α=.05 and α=.01; Q22 supplies the tyre-life power application.

9.R Recognition & Selection Lab

Identify parameter, tail and Z/t/proportion route before calculation.

No labels in the final practice lane.

9.C Calculation Lab

Execute test statistics, p-values, critical values and power.

Write hypotheses and tail first; calculate second.

9.I Interpretation Lab

Produce defensible statistical conclusions.

Do not accept H0; connect the decision to the claim and distinguish statistical from practical importance.

9.M Mixed Chapter 9 Set

Unlabelled mixture of Chapter 9 methods.

Method selection is part of the task.

9.B Chapter 9 Final Boss

Closed-source transfer across the full chapter.

A pass adds evidence but does not bypass retrieval, transfer and repair requirements.

9.X Repair

Repair the exact misconception rather than rereading everything.

Tail, Z-vs-t, p-value, proportion-condition and power errors receive targeted retests.

Formula decision objects

F6.NORMAL · $Z=(X-\mu)/\sigma$

QUESTION TYPE	Probability or boundary for normal X
OBJECT	X
SPREAD	σ
CONDITIONS	Normal model
TOOL	NORM.S.DIST / NORM.DIST; inverse variants
CHECK	Region matches wording

F6.UNIFORM · $P(c<X<d)=(d-c)/(b-a)$

QUESTION TYPE	Probability on an interval with equal density
OBJECT	X
SPREAD	$\sqrt{[(b-a)^2/12]}$
CONDITIONS	$a\leq X\leq b$ and constant density
TOOL	Arithmetic
CHECK	Probability is area, not height

F6.EXP · $P(X\leq x)=1-e^{(-\lambda x)}$

QUESTION TYPE	Waiting time between events
OBJECT	X
SPREAD	$1/\lambda$
CONDITIONS	λ and time use the same units
TOOL	Calculator
CHECK	Count would be Poisson; wait is exponential

F6.BINOM.N · Continuity-correct, then standardise

QUESTION TYPE	Approximate a binomial count with normal
OBJECT	Xa
SPREAD	$\sqrt{[n\pi(1-\pi)]}$
CONDITIONS	$n\pi>5$ and $n(1-\pi)>5$
TOOL	Calculator / Excel
CHECK	Correct ±0.5 boundary

F7.MEAN.Z · $Z=(\bar{X}-\mu)/(\sigma/\sqrt{n})$

QUESTION TYPE	Probability or boundary for sample mean when σ is known
OBJECT	$\bar{X}$
SPREAD	$\sigma/\sqrt{n}$
CONDITIONS	Normal population or justified CLT; σ known
TOOL	NORM.S.DIST / NORM.DIST
CHECK	Average, not individual

F7.MEAN.T · $T=(\bar{X}-\mu)/(s/\sqrt{n})$, $df=n-1$

QUESTION TYPE	Probability for sample mean when σ is unknown
OBJECT	$\bar{X}$
SPREAD	$s/\sqrt{n}$
CONDITIONS	σ unknown; sourced t assumptions
TOOL	T.DIST(tcalc,df,TRUE)
CHECK	Do not use Z with sample s

F7.PROP · $Z=(p-\pi)/SE$

QUESTION TYPE	Probability involving sample proportion
OBJECT	p
SPREAD	$\sqrt{[\pi(1-\pi)/n]}$
CONDITIONS	$n\pi>5$ and $n(1-\pi)>5$

TOOL	Normal CDF
CHECK	Known π versus estimated p

F7.DIFF · Z=[diff−(μ1−μ2)]/SE

QUESTION TYPE	Probability involving difference of two independent sample means
OBJECT	$\bar{X}_1-\bar{X}_2$
SPREAD	$\sqrt{(\sigma_1^2/n_1+\sigma_2^2/n_2)}$
CONDITIONS	Independent normal populations; sourced known-σ case
TOOL	Normal CDF
CHECK	Preserve group order

F8.CI.Z · Xbar +/- z(alpha/2) sigma/sqrt(n)

QUESTION TYPE	Estimate mu when population sigma is known.
OBJECT	Xbar estimates mu
SPREAD	
CONDITIONS	normal population; if not normal, OFF23 says n > 30
TOOL	Critical z; LEGACY EXCEL ROUTE
CHECK	Interpret the population mean, not an individual value.

F8.CI.T · Xbar +/- t(alpha/2,n-1) s/sqrt(n)

QUESTION TYPE	Estimate mu when population sigma is unknown.
OBJECT	Xbar estimates mu
SPREAD	
CONDITIONS	population normality support where needed; df=n-1
TOOL	Exact t may validate arithmetic; table row convention unresolved; LEGACY EXCEL ROUTE
CHECK	Claims outside the interval conflict with the estimated mean, not necessarily individual observations.

F8.CI.P · p +/- z(alpha/2) sqrt[p(1-p)/n]

QUESTION TYPE	Estimate population proportion pi.
OBJECT	p estimates pi
SPREAD	
CONDITIONS	np > 5 and n(1-p) > 5
TOOL	Critical z; LEGACY EXCEL ROUTE
CHECK	State the population proportion, interval, confidence and context.

F8.N.MEAN · n=z^2 sigma^2/e^2; ROUND UP

QUESTION TYPE	Choose n for a mean with desired margin e.
OBJECT	required sample size n
SPREAD	
CONDITIONS	large-population planning formula
TOOL	Arithmetic; LEGACY EXCEL ROUTE
CHECK	The rounded n guarantees the planned maximum margin under assumptions.

F8.N.PROP · n=z^2 p(1-p)/e^2; ROUND UP

QUESTION TYPE	Choose n for a proportion with desired margin e.
OBJECT	required sample size n
SPREAD	
CONDITIONS	use p=.5 when no useful estimate exists (OFF23-CONFIRMED)
TOOL	Arithmetic; LEGACY EXCEL ROUTE
CHECK	Explain prior p versus conservative .5 and upward rounding.

F8.FPC.N · n=n0 N/[n0+(N-1)]; ROUND UP

QUESTION TYPE	Adjust infinite-population n0 for finite N.
OBJECT	required final sample size n
SPREAD	
CONDITIONS	finite-population planning
TOOL	Arithmetic; LEGACY EXCEL ROUTE
CHECK	Final n is no greater than n0 and must be rounded upward.

F9.MEAN.Z · Z=(X̄−μ0)/(σ/√n)

QUESTION TYPE	One-sample hypothesis test for μ when population σ is known
OBJECT	$\bar{X}$
SPREAD	σ/√n
CONDITIONS	Random sample; source conditions; σ known
TOOL	Critical Z or NORM.S.DIST / NORM.S.INV
CHECK	Tail comes from H1; conclude reject/do not reject H0

F9.MEAN.T · $T=(\bar{X}-\mu_0)/(s/\sqrt{n})$

QUESTION TYPE	One-sample hypothesis test for μ when population σ is unknown
OBJECT	$\bar{X}$
SPREAD	s/√n
CONDITIONS	σ unknown; df=n−1; source-supported normality condition
TOOL	T.INV / T.DIST / T.DIST.2T
CHECK	Do not use Z merely because n is supplied

F9.PROP.Z · $Z=(p-\pi_0)/\sqrt{[\pi_0(1-\pi_0)/n]}$

QUESTION TYPE	One-sample hypothesis test for population proportion π
OBJECT	p=x/n
SPREAD	$\sqrt{[\pi_0(1-\pi_0)/n]}$
CONDITIONS	Binary categorical data; $n\pi_0\geq5$; $n(1-\pi_0)\geq5$
TOOL	Critical Z / normal CDF
CHECK	Use null π_0 in the test SE

F9.POWER · power=1−β

QUESTION TYPE	Power of a one-sample mean test at a stated true μ
OBJECT	$\bar{X}$ critical boundary
SPREAD	σ/√n
CONDITIONS	Test direction, α, σ and n supplied
TOOL	Normal inverse + normal probability
CHECK	Compute the critical $\bar{X}$ under H0 first, then β under the true μ

F8.GENERAL · estimate +/- critical value x SE

QUESTION TYPE	What is the architecture of a confidence interval?
OBJECT	sample estimate of a population parameter
SPREAD	
CONDITIONS	source conditions for the selected estimator
TOOL	Arithmetic; LEGACY EXCEL ROUTE only
CHECK	State parameter, interval, confidence, context; not individuals.

F8.FPC · sqrt[(N-n)/(N-1)]

QUESTION TYPE	Correct standard error for a finite population.
OBJECT	Xbar or p estimating mu or pi
SPREAD	
CONDITIONS	use supplied rule when n/N > .05
TOOL	Arithmetic; LEGACY EXCEL ROUTE
CHECK	FPC is below 1 and reduces uncertainty, not the centre.

F8.FINITE.MEAN · Xbar +/- t(alpha/2,n-1) (s/sqrt(n)) sqrt[(N-n)/(N-1)]

QUESTION TYPE	Estimate mu for a finite population sampled without replacement.
OBJECT	Xbar estimates mu
SPREAD	
CONDITIONS	n/N > .05 under supplied rule
TOOL	Exact t can check arithmetic; table convention unresolved
CHECK	Interpret population mean; FPC narrows the interval.

F8.FINITE.PROP · p +/- z(alpha/2) sqrt[p(1-p)/n] sqrt[(N-n)/(N-1)]

QUESTION TYPE	Estimate pi for a finite population sampled without replacement.
OBJECT	p estimates pi
SPREAD	

CONDITIONS	np > 5, n(1-p) > 5 and n/N > .05
TOOL	Arithmetic; LEGACY EXCEL ROUTE
CHECK	Interpret the finite-population proportion.

F8.BOOTSTRAP · PROCEDURE: sample -> resample with replacement -> same n -> statistic -> repeat -> order -> cut alpha/2 tails -> interval

QUESTION TYPE	Construct a percentile bootstrap interval without a false parametric formula.
OBJECT	bootstrap distribution of a statistic
SPREAD	
CONDITIONS	resample from the observed sample WITH replacement; each resample has size n
TOOL	Software is practically required
CHECK	Limits are ordered bootstrap statistics; identify the population parameter estimated.

Chapter 8 - Confidence Interval Estimation

> **PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified**
> chapterSourceStatus = SOURCE_PARTIAL; final 2026 source gate = NOT SATISFIED.

Method decision tree - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Parameter mean? -> sigma known -> z CI; sigma unknown -> Student t, s and df=n-1. Parameter proportion? -> check np>5 and n(1-p)>5 -> z CI. Required n? -> choose mean or proportion formula and ROUND UP. Finite N plus without replacement? -> inspect n/N; use FPC when n/N>.05. Ordinary parametric CI inappropriate due to non-normality? -> bootstrap.

Z versus t fork - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Cue	Route	Spread	Lock
Mean, sigma known	z	sigma/sqrt(n)	OFF23 non-normal rule: n>30
Mean, sigma unknown	Student t	s/sqrt(n)	df=n-1; non-listed table convention unresolved
Proportion	z	sqrt[p(1-p)/n]	strict np>5 and n(1-p)>5

Confidence interpretation - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

A complete report identifies the population parameter, interval, confidence level and context, and describes confidence as the long-run performance of the interval method. It does not say that a fixed parameter moves randomly after calculation and does not interpret a mean CI as a range for individuals.

Formula decisions - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified -

PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

F8.CI.Z - Xbar +/- z(alpha/2) sigma/sqrt(n)

QUESTION ANSWERED	Estimate mu when population sigma is known.
RANDOM OBJECT / PARAMETER	Xbar estimates mu
PREREQUISITES	random sample; known sigma; confidence level
CONDITIONS	normal population; if not normal, OFF23 says n > 30
CUES	mean interval; sigma known
RULED OUT BY	sigma unknown; proportion target
CONFUSED WITH	Student t CI
TOOL ROUTE	Critical z; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	Interpret the population mean, not an individual value.
COMMON TRAPS	using s; forgetting sqrt(n); using alpha instead of alpha/2
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.CI.T - Xbar +/- t(alpha/2,n-1) s/sqrt(n)

QUESTION ANSWERED	Estimate mu when population sigma is unknown.
RANDOM OBJECT / PARAMETER	Xbar estimates mu
PREREQUISITES	random sample; sample s; confidence level
CONDITIONS	population normality support where needed; df=n-1
CUES	mean interval; sigma unknown; s supplied
RULED OUT BY	known sigma; proportion target
CONFUSED WITH	known-sigma z CI
TOOL ROUTE	Exact t may validate arithmetic; table row convention unresolved; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	Claims outside the interval conflict with the estimated mean, not necessarily individual observations.
COMMON TRAPS	using z; df=n; inventing interpolation/rounding convention
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.CI.P - p +/- z(alpha/2) sqrt[p(1-p)/n]

QUESTION ANSWERED	Estimate population proportion pi.
RANDOM OBJECT / PARAMETER	p estimates pi
PREREQUISITES	success count or p; sample size n; confidence level
CONDITIONS	np > 5 and n(1-p) > 5
CUES	proportion/percentage interval
RULED OUT BY	mean target; failed approximation conditions
CONFUSED WITH	mean CI
TOOL ROUTE	Critical z; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	State the population proportion, interval, confidence and context.
COMMON TRAPS	using pi when unknown; omitting both condition checks
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.N.MEAN - n=z^2 sigma^2/e^2; ROUND UP

QUESTION ANSWERED	Choose n for a mean with desired margin e.
RANDOM OBJECT / PARAMETER	required sample size n
PREREQUISITES	confidence level; margin e; known/assumed/estimated sigma
CONDITIONS	large-population planning formula
CUES	within +/- e; mean precision
RULED OUT BY	proportion planning task
CONFUSED WITH	mean CI after data collection
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	The rounded n guarantees the planned maximum margin under assumptions.
COMMON TRAPS	rounding normally; using width instead of margin; using s without a source-supported estimate
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.N.PROP - n=z^2 p(1-p)/e^2; ROUND UP

QUESTION ANSWERED	Choose n for a proportion with desired margin e.
RANDOM OBJECT / PARAMETER	required sample size n
PREREQUISITES	confidence level; margin e; prior/pilot p or no estimate
CONDITIONS	use p=.5 when no useful estimate exists (OFF23-CONFIRMED)
CUES	proportion precision; prior estimate
RULED OUT BY	mean planning task
CONFUSED WITH	proportion CI after data collection
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	Explain prior p versus conservative .5 and upward rounding.
COMMON TRAPS	using .5 despite a useful prior estimate; rounding down
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.FPC.N - n=n0 N/[n0+(N-1)]; ROUND UP

QUESTION ANSWERED	Adjust infinite-population n0 for finite N.
RANDOM OBJECT / PARAMETER	required final sample size n
PREREQUISITES	n0; finite N
CONDITIONS	finite-population planning
CUES	finite roster; sample-size adjustment
RULED OUT BY	infinite population
CONFUSED WITH	FPC for an already collected CI
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	Final n is no greater than n0 and must be rounded upward.
COMMON TRAPS	using observed n as n0; not rounding upward
SOURCE	Chapter 8/Estimation and Sample Size Determination for Finite Populations_Beraming en Bepaling van steekproefgrootte vir eindige populasies.pdf - TXT - Section 8.7, equations 8.12-8.14

F8.GENERAL - estimate +/- critical value x SE

QUESTION ANSWERED	What is the architecture of a confidence interval?
RANDOM OBJECT / PARAMETER	sample estimate of a population parameter
PREREQUISITES	point estimator; critical distribution; standard error
CONDITIONS	source conditions for the selected estimator
CUES	estimate an unknown parameter with a range
RULED OUT BY	a probability about one observation
CONFUSED WITH	sampling distribution probability
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE only
INTERPRETATION CHECK	State parameter, interval, confidence, context; not individuals.
COMMON TRAPS	treating margin as full width; probability claim about fixed parameter
SOURCE	Chapter 8/SDS 188 chapter8_2023.pdf - OFF23 - pp. 4-71; Chapter 8 theory, worked examples, FPC and bootstrap sequence

F8.FPC - sqrt[(N-n)/(N-1)]

QUESTION ANSWERED	Correct standard error for a finite population.
RANDOM OBJECT / PARAMETER	Xbar or p estimating mu or pi
PREREQUISITES	finite N; sample n; sampling without replacement
CONDITIONS	use supplied rule when n/N > .05
CUES	finite roster; without replacement; large sampling fraction
RULED OUT BY	sampling with replacement; effectively infinite population
CONFUSED WITH	ordinary SE
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	FPC is below 1 and reduces uncertainty, not the centre.
COMMON TRAPS	using N-n over N; applying regardless of replacement
SOURCE	Chapter 8/Estimation and Sample Size Determination for Finite Populations_Beraming en Bepaling van steekproefgrootte vir eindige populasies.pdf - TXT - Section 8.7, equations 8.12-8.14

F8.FINITE.MEAN - Xbar +/- t(alpha/2,n-1) (s/sqrt(n)) sqrt[(N-n)/(N-1)]

QUESTION ANSWERED	Estimate mu for a finite population sampled without replacement.

RANDOM OBJECT / PARAMETER	Xbar estimates mu
PREREQUISITES	mean CI inputs; finite N; without replacement; FPC needed
CONDITIONS	n/N > .05 under supplied rule
CUES	finite mean interval
RULED OUT BY	replacement sampling
CONFUSED WITH	ordinary mean CI
TOOL ROUTE	Exact t can check arithmetic; table convention unresolved
INTERPRETATION CHECK	Interpret population mean; FPC narrows the interval.
COMMON TRAPS	omitting FPC; changing the centre
SOURCE	Chapter 8/Estimation and Sample Size Determination for Finite Populations_Beraming en Bepaling van steekproefgrootte vir eindige populasies.pdf - TXT - Section 8.7, equations 8.12-8.14

F8.FINITE.PROP - p +/- z(alpha/2) sqrt[p(1-p)/n] sqrt[(N-n)/(N-1)]

QUESTION ANSWERED	Estimate pi for a finite population sampled without replacement.
RANDOM OBJECT / PARAMETER	p estimates pi
PREREQUISITES	p; n; finite N; without replacement
CONDITIONS	np > 5, n(1-p) > 5 and n/N > .05
CUES	finite proportion interval
RULED OUT BY	replacement sampling
CONFUSED WITH	ordinary proportion CI
TOOL ROUTE	Arithmetic; LEGACY EXCEL ROUTE
INTERPRETATION CHECK	Interpret the finite-population proportion.
COMMON TRAPS	omitting FPC; using count instead of p
SOURCE	Chapter 8/Estimation and Sample Size Determination for Finite Populations_Beraming en Bepaling van steekproefgrootte vir eindige populasies.pdf - TXT - Section 8.7, equations 8.12-8.14

F8.BOOTSTRAP - PROCEDURE: sample -> resample with replacement -> same n -> statistic -> repeat -> order -> cut alpha/2 tails -> interval

QUESTION ANSWERED	Construct a percentile bootstrap interval without a false parametric formula.
RANDOM OBJECT / PARAMETER	bootstrap distribution of a statistic
PREREQUISITES	original random sample of size n; chosen statistic; confidence level; many resamples
CONDITIONS	resample from the observed sample WITH replacement; each resample has size n
CUES	non-normality; parametric CI inappropriate
RULED OUT BY	sampling without replacement; parametric z/t CI
CONFUSED WITH	ordinary parametric CI
TOOL ROUTE	Software is practically required
INTERPRETATION CHECK	Limits are ordered bootstrap statistics; identify the population parameter estimated.
COMMON TRAPS	resampling from the population; changing n; cutting alpha from each tail
SOURCE	Chapter 8/Bootstrapping_Skoenlusberaming.pdf - TXT - Section 8.8, steps 1-7 and 90%/100-resample example

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Sample-size rules - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified -

PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Mean: n=z^2 sigma^2/e^2. Proportion: n=z^2 p(1-p)/e^2. Use prior/pilot p when useful; otherwise p=.5 (OFF23-CONFIRMED). If sigma is unavailable, use an expected conservative sigma or pilot s (OFF23-CONFIRMED). ROUND UP every required sample size.

Finite population correction - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified -

PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Only for a finite population sampled without replacement. Inspect n/N. Use FPC=sqrt[(N-n)/(N-1)] when n/N>.05. The factor reduces SE. For planning, compute n0 then n=n0N/[n0+(N-1)] and round upward.

Bootstrap sequence - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified -

PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Population -> original random sample -> bootstrap resample FROM THE OBSERVED SAMPLE -> WITH replacement -> same n -> calculate statistic -> repeat many times -> order statistics -> cut alpha/2 tails -> interval. Source example: 90%, 100 resamples, 5th smallest=36.6296 and 5th largest=51.5926.

Tutorial 8 map and answer authority - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Item	Related skills	Answer	Authority
Q1	C8.11, C8.12, C8.13, C8.14, C8.15, C8.16	(83.0400, 86.9600)	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q2	C8.11, C8.12, C8.13, C8.14, C8.15, C8.16, C8.20	(114.6967, 135.3033)	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q3	C8.09, C8.16	95% confidence describes the long-run method; the parameter is the mean monthly sales, not individual months.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q4(a)	C8.11, C8.12, C8.13, C8.14, C8.15, C8.16	(1.9802, 1.9998) litres	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q4(b)	C8.17	No additional normal-population assumption is required under the OFF23 large-sample rule because n=100>30.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q4(c)	C8.10	2.02 litres concerns one bottle; the CI estimates the population mean and is much narrower than individual variation.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q4(d)	C8.18	(1.9602, 1.9798) litres	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q5	C8.37, C8.38, C8.39, C8.40, C8.41	35	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q6	C8.37, C8.38, C8.39, C8.40, C8.41	166	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q7(a)	C8.43, C8.44, C8.48	2377	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q7(b)	C8.43, C8.44, C8.48	1978	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q8(a)	C8.30, C8.31, C8.32, C8.33, C8.34	(0.90165, 0.95725)	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q8(b)	C8.35	We are 95% confident that the population proportion of business people whose presentations have been interrupted is between about 90.17% and 95.72%.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q8(c)	C8.43, C8.47, C8.48	158	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q8(d)	C8.43, C8.47, C8.48	272	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q9(a)	C8.22, C8.23, C8.24	2.2622	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q9(b)	C8.22, C8.23, C8.24	3.2498	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q9(c)	C8.22, C8.23, C8.24	2.0395 exact	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q9(d)	C8.22, C8.23, C8.24	1.9977 exact	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q9(e)	C8.22, C8.23, C8.24	1.7531	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q10(a)	C8.21, C8.22, C8.23, C8.24, C8.25, C8.26, C8.27	(134.6180, 147.9820)	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q10(b)	C8.28	The estimated mean interval lies below 150, so the data give evidence that the advertised mean index is not met; phrase the conclusion about the population mean, not every tyre.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution

Item	Related skills	Answer	Authority
Q10(c)	C8.29	152 is an individual tyre value; the CI estimates the population mean and is not a prediction interval for individuals.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q11	C8.49, C8.50, C8.51	(67.6282, 82.3718) exact-t check	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q12(a)	C8.49, C8.50, C8.52	(0.22846, 0.37154)	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q12(b)	C8.55, C8.56, C8.57	186	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution
Q12(c)	C8.50, C8.53, C8.54, C8.56	For N=2000: CI (0.22651, 0.37349); required n=205.	OFFICIAL_ORIGINAL stem; derived_and_independently_checked solution

t-table lock: Q9(c), Q9(d) and other non-listed df checks may show exact mathematical values; AWAIT_MEMO_OR_CURRENT_SOURCE for the lecturer table-row convention.

Common traps - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

- Known sigma -> z; unknown sigma -> t with s and df=n-1.
- Confidence concerns the interval procedure, not a probability that a fixed parameter moves.
- A mean CI is not a range for individual observations.
- Required sample size always rounds upward.
- Use prior p when useful; p=.5 only when no useful estimate exists.
- FPC requires finite N, without replacement and the supplied n/N>.05 rule.
- Bootstrap resamples from the observed sample with replacement and preserves n.
- LEGACY EXCEL ROUTE is not a 2026 EXAM PROCEDURE.

8.H Auditing / Totals - HOLD - A1 SCOPE UNCONFIRMED - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

C8.69-C8.75 remain preserved as reference-only source material and are excluded from all standard learning, evidence, mixed-set, simulation, Final Boss and readiness routing.

Source status - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified - PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified

Authority	File	Use
P26	Admin/2026 Module Framework SDS188.pdf	current assessment/tool authority
OFF23	Chapter 8/SDS 188 chapter8_2023.pdf	verified adaptation
OFF22	Chapter 8/SDS 188 Tutorial 8_2022.pdf	OFFICIAL_ORIGINAL
OFF	Chapter 8/SDS 188 CHAPTER 8 t-table.pdf	critical-value reference
TXT	Chapter 8/Estimation and Sample Size Determination for Finite Populations_Beraming en Bepaling van steekproefgrootte vir eindige populasies.pdf	VERIFIED_ADAPTATION
TXT	Chapter 8/Bootstrapping_Skoenlusberaming.pdf	VERIFIED_ADAPTATION
TXT/HOLD	Chapter 8/Confidence Intervals for Totals_Vertrouensintervalle vir totale.pdf	HOLD_ONLY_REFERENCE

A1S2 2026 Assessment-Execution Supplement

> **Assessment-training purpose.**
> The 2026 practice test demonstrates an EMSLearn + Excel workflow. Train by reading the EMSLearn-style prompt, working in Excel where required, then entering the exact requested response. Practice-test numerical answers use exactly four decimals, a full stop, and no units/currency symbols.

> **Assessment logistics source.**
> The EMSLearn **"A1S2 Practice test" announcement** (opened Monday, 17 August 2026, 4:00 PM) states: "Similar rules and instructions will be provided in the A1S2" and explains that the practice quiz is password protected "just as the quiz will require a password during the A1S2." The practice-test password itself is `maties`; this does not imply that the real A1S2 will use the same password.

Practice-test blueprint

Practice Q	Chapter	Observed skill/execution family
1	Ch 8	**Bootstrapping** Use the Bootstrap worksheet; distinguish the original sample from with-replacement resamples and obtain bootstrap CI limits.
2	Ch 8	**Proportion confidence interval** Compute p and its estimated SE; notice that a later EMSLearn sub-part may supply fresh values that override earlier calculations.
3	Ch 9	**Critical value + decision** Choose the correct tail/distribution, compute a critical value and compare the test statistic.
4	Ch 9	**Raw-data mean test + power** Use the calls worksheet; convert the critical point to original units, calculate power and reason about Type II error.
5	Ch 8	**Proportion sample size** With no prior proportion estimate, use p=0.5 and round the required sample size upward.
6	Ch 8	**Finite-population sample size** Calculate n_0, then adjust for finite N when sampling without replacement.
7	Ch 9	**Type II error** Recognise "do not reject H_0 when H_0 is false" in context.
8	Ch 7/8	**Unknown σ / Student t sampling mean** Use s, df=n−1 and a t probability for a sample-mean interval.

Formula-sheet reality check

Area	Given on 2026 sheet	You still have to know / execute
Chapter 6 distributions	Uniform and exponential density/mean/variance formulas are supplied.	Choose the distribution, translate the event, use complement/inverse logic and execute the correct Excel function.
Sample-size determination	Mean, proportion and finite-population sample-size formulas are supplied.	Choose z from confidence, choose p or conservative 0.5, apply finite-population adjustment only when appropriate and round required n upward.
Mean inference	Rows distinguish σ known → Z and σ unknown → Student t, with the relevant standard error.	Recognise the route, determine df for t, choose tails/critical probabilities, transform critical statistics back to original units and interpret.
Proportion inference	The π / p standard-error structure is supplied.	Compute p, choose CI/test context, check conditions and distinguish theoretical from estimated standard error.
Two independent means	The $\mu_1-\mu_2$ statistic, standard error $\sqrt{(\sigma_1^2/n_1+\sigma_2^2/n_2)}$, and Z route when σ_1 and σ_2 are known are supplied.	Preserve group order, check independence/model conditions, translate the event, standardise and interpret the difference.
Hypothesis-test reasoning	A complete test-decision algorithm is not supplied.	Know H_0/H_1 direction, Type I/II, p-value/critical-value decision language, β/power and practical interpretation.
Bootstrapping	No bootstrap algorithm appears on the one-page formula sheet.	Recognise when bootstrap is appropriate and execute original sample → with-replacement resamples → statistic → repeat → ordered distribution → interval logic.

Altered-number Excel execution drills

Work these using the companion workbook where indicated. These are training analogues, not copied assessment questions.

EX01 · ALTERED-NUMBER EXECUTION DRILL

Bootstrap execution from Excel

Workbook: Bootstrap_Practice

In M5 calculate the mean of the first resample and fill down to M204. For this altered training drill, use the 5th smallest and 5th largest resample means as the 95% training limits.

Route: =AVERAGE(C5:L5) → fill down → =SMALL(M5:M204,5) and =LARGE(M5:M204,5).

Lower: _____ Upper: _____

EX02 · ALTERED-NUMBER EXECUTION DRILL

Population-proportion confidence interval

No workbook required

Parts (a)–(b): n=420, x=273. Compute p and its estimated SE. Part (c) is deliberately decoupled, matching the EMSLearn practice-test mechanic: ignore your earlier values and use supplied p=0.62 and SE=0.0240 to find the upper 95% confidence limit.

Route: p=x/n; SE=SQRT(p*(1-p)/n). For part (c): 0.62 + NORM.S.INV(0.975)*0.0240.

p: _____ SE: _____ Supplied-value upper limit: _____

EX03 · ALTERED-NUMBER EXECUTION DRILL

Student-t critical value and decision

No workbook required

H0: μ=9.2 versus H1: μ>9.2; σ unknown; n=20; α=0.05; tcalc=2.05.

Route: df=19; =T.INV(0.95,19); reject when tcalc>tcrit.

Critical t: _____ Decision: _____

EX04 · ALTERED-NUMBER EXECUTION DRILL

Raw-data known-σ mean test + power

Workbook: Calls_Practice

H0: μ=12 versus H1: μ>12 at α=0.01, σ=4.5, n=36. Use the raw calls data. Also calculate power if the true mean is 14.

Route: COUNT + AVERAGE; z*=NORM.S.INV(0.99); critical $\bar{X}$=12+z*4.5/SQRT(n); zcalc=($\bar{X}$–12)/(4.5/SQRT(n)); power=1–NORM.DIST(critical $\bar{X}$,14,4.5/SQRT(n),TRUE).

Critical boundary: _____ zcalc: _____ Power: _____

EX05 · ALTERED-NUMBER EXECUTION DRILL

Required n with no prior proportion estimate

No workbook required

95% confidence, margin of error ±0.055, no prior π estimate. Determine the minimum n.

Route: use p=0.5; $n_0=z^2p(1-p)/e^2$; ROUND UP.

Minimum n: _____

EX06 · ALTERED-NUMBER EXECUTION DRILL

Finite-population sample-size adjustment

No workbook required

N=2400, margin ±0.025, 99% confidence, pilot p=0.42, sampling without replacement.

Route: $n_0=z^2p(1-p)/e^2$; then $n=n_0N/[n_0+(N-1)]$; ROUND UP.

Adjusted n: _____

EX07 · ALTERED-NUMBER EXECUTION DRILL

Recognise Type II error

No workbook required

H_0 says the population mean weekly study time is 18 hours; H_1 says it differs from 18. Which outcome is a Type II error?

Route: Type II = do not reject H_0 when H_0 is false.

Answer: _____

EX08 · ALTERED-NUMBER EXECUTION DRILL

Unknown-σ sample-mean probability with Student t

No workbook required

Centre=22 apps/month, σ unknown, n=30, s=5. Find $P(21.5<\bar{X}<22.5)$ using Student t.

Route: SE=s/SQRT(n); df=29; compute tL,tU; =T.DIST(tU,29,TRUE)-T.DIST(tL,29,TRUE).

Probability: _____

Chapter 8 evidence state: the overall canonical gate remains SOURCE_PARTIAL. Skills directly represented in the 2026 practice test may be tagged PRACTICE_CONFIRMED_2026, while auditing/population-total C8.69–C8.75 remain HOLD.

Source register

ENGINE v1.3.1 + CHAPTER 8 PROVISIONAL CONTENT PACK
PROVISIONAL CORE - Current 2026 Chapter 8 package not yet fully verified. C8.01-C8.68 PROVISIONAL_CORE; C8.69-C8.75 HOLD; final readiness gate remains closed.

CHAPTER 9 · SOURCE_READY
SDS chapter9_2023.pdf · P26-REUSED · official 2023 deck confirmed as the official 2026 deck · pp. 1–79. SDS Tutorial 9.pdf is the official practice spine. Tutorial Q15–16 and Q19–20 are classified as Chapter 8 confidence-interval / population-total work and are excluded from Chapter 9 mastery.

CHAPTER 8 CURRENT-TRANSCRIPT SUPPORT — STATUS LOCK PRESERVED
Transcripts/SDS TRANSCRIPT 10.rtfd/TXT.rtf · P26 · 2026 · 5 August 2026 · 2:01–4:00 — supports C8.11, C8.21, C8.22, C8.30, C8.31, C8.32, C8.33, C8.34
Transcripts/SDS TRANSCRIPT 11.rtfd/TXT.rtf · P26 · 2026 · 7 August 2026 · 3:53–9:50 — supports C8.19, C8.20, C8.34, C8.37, C8.40, C8.43, C8.44, C8.45, C8.46, C8.47, C8.48
C8.45: current Transcript 11 confirms p=.5 when no prior estimate exists because it maximises p(1−p) and gives a conservative larger sample. Chapter 8 remains SOURCE_PARTIAL / PROVISIONAL. Chapter 9 is SOURCE_READY from the supplied official 2023 deck confirmed as current for 2026; its Tutorial 9 practice scope is tracked separately.

v1.3.1 FREEZE METRICS
Canonical 223 · ACTIVE 141 · PROVISIONAL 75 · HOLD 7 · tasks 1577 · multi-assessed tasks 9. Manifest PASS; changed variants 141/141; source-integrity issues 7. Pathways: UNDERSTAND 141/141; genuine RETRIEVE 141/141; genuine APPLY 141/141; changed transfer 141/141; delayed 141/141.
Task/evidence counts: simple choice 308; production-first choice 120; numeric 207; formula 20; symbolic 25; region 12; structured interpretation 104; Excel reconstruction 25; Excel independent execution 1; short text 598; ordered steps 1; multi-part production 179.

CORRECTED CURRENT SOURCE FINDING
SDS Chapter 7 NOTES.pdf · P26 · 2026 · §2.6, p. 7 directly verifies Student-t, T, df=n−1 and T.DIST. 7/7 Student-t skills are ACTIVE. Chapter 7 current core is SOURCE_READY; C7.52-C7.58 FPC remain separately PROVISIONAL / legacy-supported.

Evidence contract: skillIds route related topics. Only assessesSkillIds award mastery and appear in the semantic audit. **Source-integrity issues:** 7.

Loaded: 223 skills (ACTIVE 141, PROVISIONAL 75, HOLD 7) · 1577 tasks/variants (ACTIVE 999, PROVISIONAL 527, HOLD 51).

Chapter	Total	Official	Verified adaptation	Generated validated	ACTIVE	PROVISIONAL	HOLD	Numeric	Formula	Symbolic	Region	Excel	Interpretatic
6	365	4	24	286	365	0	0	57	2	0	12	8	9
7	356	0	20	285	355	1	0	40	5	10	0	8	6
8	577	30	360	0	0	526	51	58	8	0	0	8	10
9	279	0	37	200	279	0	0	52	5	15	0	5	0

Chapter	File	Year	Authority	Status
6	Chapter 6.pdf	2026	P26	SOURCE_READY
7	SDS Chapter 7 NOTES.pdf	2026	P26	SOURCE_READY
8	SDS 188 chapter8_2023.pdf	2023	OFF23	SOURCE_PARTIAL
9	SDS chapter9_2023.pdf · official deck reused in 2026	2023	P26-REUSED	SOURCE_READY
6	Tutorial 6.pdf + Tutorial 6 Solutions.pdf	2026	P26	OFFICIAL_PRACTICE
6	Chapter 6 Excel examples + solutions; T06-data.xlsx	2026	P26	OFFICIAL_TOOL_LANE
7	Tutorial 7_2023.pdf	2023	OFF	LEGACY_OFFICIAL_PRACTICE
7	SDS188_Chapter7_Sampling_Distributions_MasterStudyManual_v1_1_FINAL.html	2026	GEN	CONTENT_DONOR_ONLY
8	SDS 188 Tutorial 8_2022.pdf	2022	OFF22	PROVISIONAL_PRACTICE
9	SDS Tutorial 9.pdf	—	OFF	OFFICIAL_PRACTICE

Chapter	File	Year	Authority	Status
8	Estimation and Sample Size Determination for Finite Populations	2021	TXT	PROVISIONAL_SUPPORT
8	Bootstrapping_Skoenlusberaming.pdf	2021	TXT	PROVISIONAL_SUPPORT
8	Confidence Intervals for Totals	2021	TXT/HOLD	HOLD
8	Transcripts/SDS TRANSCRIPT 10.rtfd/TXT.rtf	2026	P26	CURRENT_TRANSCRIPT_SUPPORT
8	Transcripts/SDS TRANSCRIPT 11.rtfd/TXT.rtf	2026	P26	CURRENT_TRANSCRIPT_SUPPORT

▶ Question-level source traceability

▶ Semantic evidence coverage by ACTIVE skill (141/141 PASS)

▶ Question mastery-target register

▶ Canonical evidence-response contract (1625/1625 task-requirement checks PASS)